

Speed Profile Tracking by an Adaptive Controller for Subway Train Based on Neural Network and PID Algorithm

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Abstract—Train running performance largely depends on the controller. In order to improve the controller performance of speed profile tracking, an adaptive control method for subway trains is addressed in this paper. Considered time-varying parameters of train motion, the train model with dynamic parameters is constructed. The model free adaptive controller combined with neural network and proportional-integral-derivative (PID) algorithm is proposed to realize adaptive control. The control laws and parameter tuning laws of the proposed controller are both illustrated, and PID gain range and actuator saturation are taken into consideration. Compared with other controllers, numerical tests are provided to verify the effectiveness and stability of the proposed controller. The controller is also applied on a real line, and the results show that the proposed controller outperforms the currently used controller. More specifically, the proposed controller reduces average absolute running time error by 0.67 s, average absolute stopping position error by 0.09 m, average maximum absolute speed tracking error by 1.22 m/s, and total energy consumption by 2.73 kWh.

Index Terms—Subway train control, model free adaptive control (MFAC), neural network, proportional-integral-derivative (PID) control, actuator saturation.

I. INTRODUCTION

SUBWAY system plays an important part of urban transportation, and the performance of train running control largely effects the passenger experience. Many researches focus on train cooperation control, thus, single train running control is a basic and vital issue [1], [2]. The performance of single train running control has significant effects on running time, energy consumption, stopping accuracy, etc. [3].

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The subway train is a servo system, and operation optimization of the single train mainly consists of two vital issues, which are speed profile optimal design and speed profile tracking control [4]. The speed profile is designed off-line, and the train controller tracks it online to make sure the nearly same performance between design and practice. The crux of speed profile tracking is developing an efficient controller.

In many studies about train tracking control, all or most parameters of train system models are constant [5], [6], even though they are various under different situations in practice. Constant parameters imply that system dynamic is not formulated well, and may cause inaccurate conclusions. Inspired by the piecewise train model of Mao *et al.* [7], the train running model with dynamic descriptions is developed in this study. The parameters such as train traction/braking force under different speed, basic resistance of different running position, train mass of different running section, gradient of different running position, and running disturbance of different running time are detailed illustrated.

As for speed profile tracking control method, proportional-integral-derivative (PID) control is the most widely used one in industrial process control [8]. PID controller controls the system through the linear combination of proportion, integration and differentiation of the control error. Many industrial controllers are implemented based around PID algorithm, as the simple structure, good robustness, and ease of use offered by PID controller [9]. PID controller is also widely used in the train running control [10]. However, it is not precise enough for some nonlinear system with dynamic parameters [11]. Hence, many researchers focus on adaptive control methods for train speed profile tracking.

There are many studies of adaptive tracking control for single train, and fuzzy logic is one of the most used algorithms. Fuzzy control is firstly added in automatic train operation (ATO) system for speed profile tracking of single train by Yasunobu *et al.* [12], and the controller effectiveness is validated through Sendai metro. Yang *et al.* [13] proposed a kind of T-S fuzzy model to identify the train model parameters, and further combined it with predictive controller to realize adaptive tracking control of the train. Moaveni *et al.* [14] also combined fuzzy logic with predictive control to track speed profile with the prevention of wheel slip. Pu *et al.* [15] proposed an integrated

optimal method for train running through speed profile design and fuzzy PID (FPID) controller construction. Combined with prediction part, Zhu *et al.* [16] also controlled train running through FPID controller, and the effectiveness of the method is validated through simulation.

Neuron network is also considered in many researches of adaptive train tracking control. Kumarawadu and Lee [17] proposed a radial basis function neural network (RBFNN) adaptive model-based vehicle control algorithm, and the controller is verified through simulations under uncertainties and severe driving conditions. Chen *et al.* [18] used single-neuron PID controller of ATO system to control the train running, and simulation shows better performance compared with PID controller. Gao *et al.* [19] introduced neural network to approximate unknown dynamics caused by actuator failure, speed profile tracking results from simulation verify the effectiveness of controller under uncertain actuator failures. Similarity, Li *et al.* [20] also used nonlinearly layered neural network to deal with fault tolerant control of train speed profile tracking. Yang *et al.* [21] constructed a multi-model controller, which includes neural network controller and FPID controller, to make train tracking meet the requirements, and relevant control methods are compared.

Some other adaptive controllers are also applied on the train. Akram *et al.* [11] established an online adaptive PID controller, which uses gradient descent algorithm to adjust PID gains, and controller performances under different disturbances are studied. Huang *et al.* [22] proposed an adaptive iterative learning control (ILC) scheme for train speed profile tracking to handle system uncertainties, simulation results are improved with iteration number increases. Aradi *et al.* [23] proposed a kind of nonlinear model predictive control (NMPC) method, and this method controls train running online with the consideration of energy consumption.

The controllers of most studies mentioned above are based on system characteristic analysis or with complicated structure to adjust parameters. And some of these controllers have obviously different structure with the currently used PID controller, which may cause big work of controller substituting. The exact model of the plant is hardly available for complicated nonlinear system, therefore, an adaptive PID-based controller without identification model is necessary. As illustrated before, fuzzy logic and neural network are two mainly techniques to adjust PID gains. The fuzzy logic needs to be designed carefully for the instability of nonlinear plant [24], while neural network has an advantage to solve such problem owing to its learning and adaptive abilities [25]. Therefore, neural network with an online learning framework is utilized to adjust PID gains in this study.

Unexpected faults are ubiquitous during train running [26], the methods of fault detection and diagnosis (FDD) techniques for train are summarized by Chen and Jiang [27]. Saturation is a common phenomenon of system due to the requirements of safety and performance, meanwhile, saturation may also bring system instability [28]. For the constraints of train traction/braking system, it's not appropriate to apply force of arbitrary value on the train. Many studies mentioned above do not consider the traction/braking system saturation, while this study takes into account relevant problems.

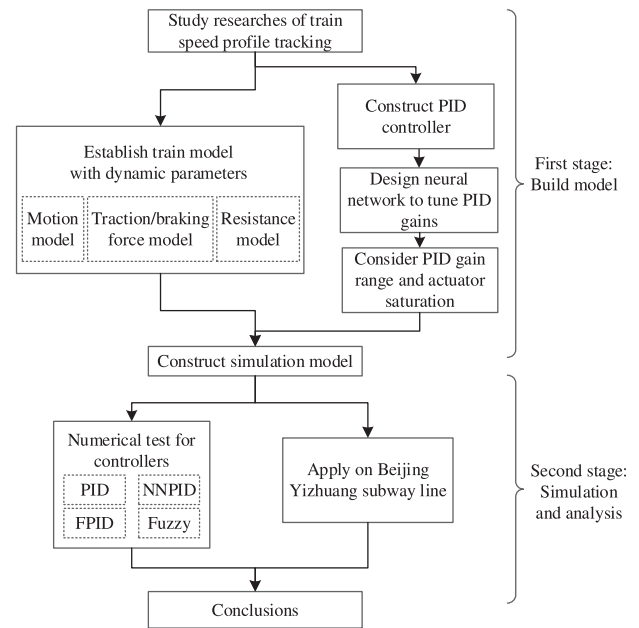


Fig. 1. Flow chart for the adaptive controller design of subway train.

This paper aims to design a novel adaptive controller that can improve the train tracking performance, the flow chart of study is shown as Fig. 1. In the first study stage, to take the advantages of relevant control methods, an adaptive controller is proposed based on neural network and PID algorithm, which shorted as NNPID controller. The train model with dynamic parameters is also constructed for the controller. In the second study stage, simulations are made. The proposed controller can track speed profile with better performance compared with several controllers under the same conditions. And in the application of Beijing Yizhuang line, nearly all indicators are improved.

The main contributions of this paper are summarized as follows:

- 1) Considered with dynamic situations of practice, train model with dynamic parameters is constructed to describe the train longitudinal motion.
- 2) A novel adaptive controller without identification model is proposed in this study to take the advantages of neural network and PID algorithm, and the controller is effective with simple structure. The controller structure and parameter on-line laws are both detailed illustrated.
- 3) The actuator saturation scheme and the PID gain range are both studied. The actuator saturation of traction/braking system is considered in the proposed controller, and corresponding control strategy is described. The signs of PID gains are invariant through the introduction of Sigmod function.

The remainder of this paper is organized as follows. Section II describes the subway train model with dynamic parameters, and the model is used as simulation plant. Section III presents control laws and control parameter tuning laws of the proposed controller, PID gain sign and actuator saturation are also analyzed in this section. Section IV shows numerical tests and the application of Beijing subway Yizhuang line, relevant results are analyzed. Section VI concludes this study.

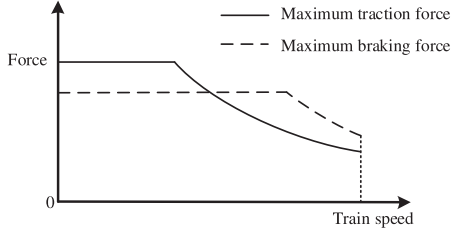


Fig. 2. Train typical maximum traction and braking characteristic curves.

II. PROBLEM FORMULATION WITH DYNAMIC PARAMETERS

This section gives a detailed continuous-time train running model. The control signal, traction/braking force, resistance, and dynamic environment parameters are analyzed in this section.

A. Train Motion Equation

Train motion equation illustrates the relationship between train running states and environments. Train is usually dealt as a mass point in the analysis of motion along the track [29], [30]. According to Newton's law, train motion equation is described as (1) in longitudinal direction:

$$\tilde{M}(t)\ddot{x}(t) = F(t) - R(t) \quad (1)$$

$$\tilde{M}(t) = (1 + \gamma)M(t) \quad (2)$$

where t is the running time, $\tilde{M}(t)$ represents the train rotating mass, $\ddot{x}(t)$ is the train acceleration value, $F(t)$ is the traction/braking force of the train, and $R(t)$ is the resistance acts on the train. The train mass doesn't match the rest mass in the running, this phenomenon is caused by the moment of inertia, and train mass is revised as rotating mass in dynamic calculation. Equation (2) shows the calculation formula of rotating mass. γ is rotating mass coefficient, and $M(t)$ is the total rest mass of the train, which consists of vehicle mass and passenger mass. $x(t)$ is train running position, train speed $\dot{x}(t)$ is the derivative of $x(t)$ versus time, and train acceleration $\ddot{x}(t)$ is the derivative of $\dot{x}(t)$ versus time. The traction/braking force $F(t)$ is generated through the traction/braking system. The train resistance $R(t)$ is influenced by train characteristics and environment conditions.

Remark 1: The variables in (1) are time-variant, specifically, relevant parameters are decided through different running circumstances and correlated with running time.

B. Traction/Braking Force

Train running is driven by the motor through traction/braking control. The train typical characteristic curves of traction/braking force are shown in Fig. 2. The maximum traction force and the maximum braking force that motor output are correlated to the train running speed. Usually, the constant stage and the change stage are both existed in traction/braking characteristic curve. With a specific train speed, the maximum traction/braking force can be obtained through (3):

$$\begin{cases} F_{a\max}(t) = f_{a\max}(\dot{x}(t)) \\ F_{b\max}(t) = f_{b\max}(\dot{x}(t)) \end{cases} \quad (3)$$

where $F_{a\max}(t)$ is the maximum traction force, $F_{b\max}(t)$ is the maximum braking force, and $f_{a\max}(\dot{x}(t))$ is the function describes the relationship between the maximum traction force and the train speed, while $f_{b\max}(\dot{x}(t))$ describes the relationship between the maximum braking force and the train speed.

As many researches illustrate [31], ATO system decides the signal of force output percent. Hence, the output percent of the maximum traction/braking force is used as the train control signal $u(t)$. And the train traction/braking force $F(t)$ can be obtained through following formula:

$$F(t) = u(t)F_{\max}(t) \\ = \begin{cases} u(t)f_{a\max}(\dot{x}(t)), & \Xi_1(t) = 1, \Xi_2(t) = 0 \\ 0, & \Xi_1(t) = 0, \Xi_2(t) = 0 \\ u(t)f_{b\max}(\dot{x}(t)), & \Xi_1(t) = 0, \Xi_2(t) = 1 \end{cases} \quad (4)$$

where $F_{\max}(t)$ is the maximum traction/braking force, $u(t)$ is the percent of the maximum traction/braking force, $|u(t)| \leq 1$, $\Xi_1(t)$ and $\Xi_2(t)$ are command indicators.

Remark 2: As Fig. 2 shown, traction force and braking force are positive. Considering opposite direction of traction and braking force, the sign is introduced for $u(t)$. $u(t)$ with positive sign is the output percent of traction force, and $u(t)$ with negative sign is the output percent of braking force. To indicate which characteristic curve should be employed, indicators $\Xi_1(t)$ and $\Xi_2(t)$ are utilized. Traction characteristic curve is used when $\Xi_1(t) = 1$, $\Xi_2(t) = 0$, which means traction command. Braking characteristic curve is used under braking command with $\Xi_1(t) = 0$, $\Xi_2(t) = 1$. Both $\Xi_1(t)$ and $\Xi_2(t)$ equal to 0 means no characteristic curve is used. Note that, to avoid command contradiction, it is not permitted that $\Xi_1(t)$ and $\Xi_2(t)$ equal to 1 at the same time.

C. Resistance Force

Train resistance plays an important part of train running. Basic resistance, gradient resistance and uncertain disturbance are considered as (5) [19]:

$$R(t) = R_b(t) + R_g(t) + \tilde{\varepsilon}(t) \quad (5)$$

where $R_b(t)$ is basic resistance, $R_g(t)$ is gradient resistance, and $\tilde{\varepsilon}(t)$ is uncertain disturbance force.

Basic resistance is usually described through a second-order polynomial of speed named as Davis formula as following [30]:

$$R_b(t) = a(t) + b(t)\dot{x}(t) + c(t)\dot{x}^2(t) \quad (6)$$

here, $a(t)$, $b(t)$, and $c(t)$, are empirical parameters. Basic resistance is mainly caused by mechanical resistance and aerodynamic resistance. In this study, subway train running on the ground as well as train running underground are considered with different basic resistance parameters, thus the parameters of basic resistance can be described in piecewise way.

Gradient resistance is caused by track gradient and can be calculated through:

$$R_g(t) = M(t)g \sin(\arctan(\iota(t))) \approx M(t)g\iota(t) \quad (7)$$

where g is gravity, and $\iota(t)$ is the gradient. The approximation in (7) is obtained according to the Taylor expansion of $\sin$ and $\tan$ with small $\iota(t)$ (track construction requirement).

To provide system uncertainty, disturbance force of train running is involved as $\tilde{\varepsilon}(t)$. Reconstruction error is also made up largely through this dynamic parameter.

Remark 3: As illustrated before, traction force always has the same direction of train running, and braking force is opposite. As for resistances, basic resistance is always in the opposite running direction, gravity resistance direction is decided by the sign of gradient, and the direction of disturbance force is uncertain. The uncertain directions of gravity resistance and disturbance force mean these two forces may benefit train running. For example, if train is running on a downward track with traction command, the gravity resistance will have the same direction of traction force which can enhance the traction effect.

D. Dynamic Train Model

Based on the analyses above, (1) can be rewritten as (8):

$$(1 + \gamma)M(t)\ddot{x}(t) = u(t)F_{\max}(t) - (a(t) + b(t)\dot{x}(t) + c(t)\dot{x}^2(t)) - M(t)g\iota(t) - \tilde{\varepsilon}(t) \quad (8)$$

To concise the calculation of train output force and basic resistance, vectors are used:

$$(1 + \gamma)M(t)\ddot{x}(t) = u(t)\mathbf{\Gamma}(t)\mathbf{\Xi}(t) - \mathbf{H}(t)\mathbf{\Omega}(t) - M(t)g\iota(t) - \tilde{\varepsilon}(t) \quad (9)$$

$\mathbf{\Gamma}(t) \in \mathbb{R}^2$ is the maximum train traction/braking force, $\mathbf{\Xi}(t) \in \mathbb{R}^2$ indicates the train running command correlated with traction/braking, and $\mathbf{\Xi}(t)$ is a column vector with two binary elements. $\mathbf{H}(t) \in \mathbb{R}^3$ contains dynamic parameters of basic resistance. $\mathbf{\Omega}(t) \in \mathbb{R}^3$ is combined with constant 1, liner term and quadratic term of the train speed. $\mathbf{\Gamma}(t)$, $\mathbf{\Xi}(t)$, $\mathbf{H}(t)$, and $\mathbf{\Omega}(t)$ can be further described as:

$$\mathbf{\Gamma}(t) = [f_{a\max}(\dot{x}(t)), f_{b\max}(\dot{x}(t))] \quad (10)$$

$$\mathbf{\Xi}(t) = [\Xi_1(t), \Xi_2(t)]^T \quad (11)$$

$$\mathbf{H}(t) = [a(t), b(t), c(t)] \quad (12)$$

$$\mathbf{\Omega}(t) = [1, \dot{x}(t), \dot{x}^2(t)]^T \quad (13)$$

with

$$u(t) \in [-1, 1], \Xi_i(t) \in \{0, 1\}, \sum_{i=1}^2 \Xi_i(t) \leq 1 \quad (14)$$

Train moves under the resultant force, and our model also updates train states through the resultant force, and the train acceleration is obtained through (15):

$$\ddot{x}(t) = \frac{1}{1 + \gamma} (u(t)\mathbf{\Gamma}_u(t)\mathbf{\Xi}(t) + \mathbf{H}_u(t)\mathbf{\Omega}(t) + g\iota(t) + \tilde{\varepsilon}_u(t)) \quad (15)$$

with

$$\mathbf{\Gamma}_u(t) = \frac{\mathbf{\Gamma}(t)}{M(t)} \quad (16)$$

$$\mathbf{H}_u(t) = \frac{\mathbf{H}(t)}{M(t)} = [a_u(t), b_u(t), c_u(t)] \quad (17)$$

$$\tilde{\varepsilon}_u(t) = \frac{\tilde{\varepsilon}(t)}{M(t)} \quad (18)$$

here, $\mathbf{\Gamma}_u(t) \in \mathbb{R}^2$ is the train maximum traction/braking force of unit mass, $\mathbf{H}_u(t) \in \mathbb{R}^3$ is the vector contains basic resistance parameters of unit train mass, and $\tilde{\varepsilon}_u(t)$ is estimated uncertain disturbance force of unit mass.

Remark 4: The train model with dynamic parameters constructed in this section is utilized as the simulation plant of train control. It also can be seen that train is a nonlinear system. Nearly all parameters of the model in this study are considered dynamically, and can be described with dynamic parameters of different conditions. Train static mass $M(t)$ various with different running sections. The maximum traction/braking force $\mathbf{\Gamma}(t)$ is correlated with train speed $\dot{x}(t)$ at time t . Train running command indicator $\mathbf{\Xi}(t)$ is various according to the train control command. $\mathbf{H}_u(t)$ is decided by the train running conditions correlated with train position $x(t)$. Vector $\mathbf{\Omega}(t)$ is correlated with train running speed $\dot{x}(t)$. Track gradient $\iota(t)$ is decided by the train position $x(t)$. Estimated uncertain disturbance force of unit mass $\tilde{\varepsilon}_u(t)$ is correlated with train running time t .

III. NNPID CONTROLLER WITH ACTUATOR SATURATION

To make better control effect in adaptiveness and robustness, an adaptive controller without identification model based on neural network and PID algorithm is construed in this section. The control algorithm is described in discrete-time to accord with ATO control cycle.

A. NNPID Controller Design

PID controller is widely used in industry and engineering for its simple structure and robustness. The continuous-time control logic of PID controller with saturation is described as:

$$u(t) = \text{sat}(\underline{U}, U(t), \bar{U}) = \begin{cases} \underline{U}, & U(t) < \underline{U} \\ U(t), & \underline{U} \leq U(t) \leq \bar{U} \\ \bar{U}, & U(t) > \bar{U} \end{cases} \quad (19)$$

$$U(t) = K_p(t)e(t) + K_i(t) \int_0^t e(t)dt + K_d(t) \frac{de(t)}{dt} \quad (20)$$

$$e(t) = r(t) - y(t) \quad (21)$$

where, $U(t)$ is the output signal of the controller at time t , $u(t)$ is the input signal of the plant at time t , $\underline{U}$ is the lower bound of the actuator output, $\bar{U}$ is the upper bound of the actuator output, $K_p(t)$ is the proportional gain, $K_i(t)$ is the integral gain, $K_d(t)$ is the differential gain, and $e(t)$ is the system error at time t and it can be calculated through (21). $r(t)$ is the reference output of the plant and $y(t)$ is the actual output of the plant.

Formulas above are described in continuous-time, whereas, some controllers are discrete. The following approximations can be made when the sampling period is small enough:

$$k = \left\lceil \frac{t}{\Delta t} \right\rceil \quad (22)$$

$$\int_0^t e(t)dt \approx \sum_{i=0}^k e(i)\Delta t \quad (23)$$

$$\frac{de(t)}{dt} \approx \frac{e(k) - e(k-1)}{\Delta t} \quad (24)$$

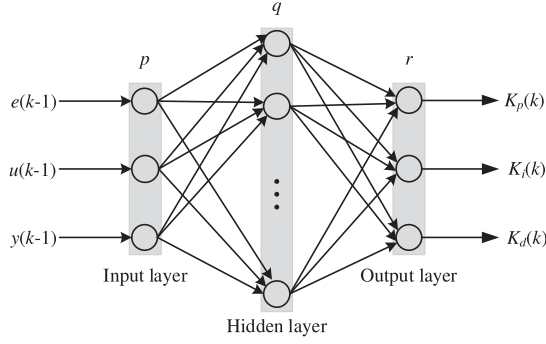


Fig. 3. Proposed neural network structure.

in which k is the sampling sequence number, Δt is sampling period. Eq. (19), (20), and. (21) are rewritten as:

$$u(k) = \text{sat}(U, U(k), \bar{U}) = \begin{cases} \underline{U}, & U(k) \leq \underline{U} \\ U(k), & \underline{U} < U(k) < \bar{U} \\ \bar{U}, & U(k) \geq \bar{U} \end{cases} \quad (25)$$

$$U(k) = K_p(k)e(k) + K_i(k)\Delta t \sum_{i=0}^k e(i) + K_d(k) \frac{e(k) - e(k-1)}{\Delta t} \quad (26)$$

$$e(k) = r(k) - y(k) \quad (27)$$

Remark 5: The ATO system outputs control signal periodically, and the sampling period is decided by the relevant design standard. In order to match the train control period, adaptive gains are also adjusted in each period in this study.

To adaptively adjust the PID gains, a three-layer neural network is constructed as Fig. 3 shown. The inputs of the proposed neural network are tracking error $e(k-1)$, $u(k-1)$, and plant output $y(k-1)$, which represented by $I_1(k)$, $I_2(k)$, and $I_3(k)$ and combined as vector $\mathbf{I}(k) \in \mathbb{R}^3$ as (28). The outputs are PID gains $K_p(k)$, $K_i(k)$, and $K_d(k)$, which can be noted as $K_1(k)$, $K_2(k)$, and $K_3(k)$ and combined as vector $\mathbf{K}(k) \in \mathbb{R}^3$ as (29).

$$\begin{aligned} \mathbf{I}(k) &= [I_1(k), I_2(k), I_3(k)]^T \\ &= [e(k-1), u(k-1), y(k-1)]^T \end{aligned} \quad (28)$$

$$\mathbf{K}(k) = [K_1(k), K_2(k), K_3(k)]^T = [K_p(k), K_i(k), K_d(k)]^T \quad (29)$$

Remark 6: The neuron number of the hidden layer can be adjusted properly for different applications to realize better control effects. In this study, there are five neurons of the hidden layer, which means $q = 5$. And the neuron number of the hidden layer is chosen based on experience and tests.

In the proposed neural network, p , q , r are used to denote each neuron of the input layer, the hidden layer, and the output layer respectively for convenience sake. In the formulas (30) of the input layer, $I_p(k)$ denotes the element of $\mathbf{I}(k)$, $N_p^{(1)}(k)$ is the input of each neuron in the input layer, $O_p^{(1)}(k)$ is the output of each neuron in the input layer, $f^{(1)}$ is the active function of the

input layer, and $\mathbf{O}^{(1)}(k) \in \mathbb{R}^3$ is the output vector of the input layer. As for the hidden layer expressed as (31), $N_q^{(2)}(k)$ is the input of each neuron in the hidden layer, $O_q^{(2)}(k)$ is the output of each neuron in the hidden layer, $f^{(2)}$ is the active function of the hidden layer, and Gaussian function is chosen as the active function of radial basis function, where $\mathbf{C}_q(k-1) \in \mathbb{R}^3$ denotes the center vector of each neuron in the hidden layer as (33) expressed, $\|N_q^{(2)}(k) - \mathbf{C}_q(k-1)\|$ is the Euclidean distance between $N_q^{(2)}(k)$ and $\mathbf{C}_q(k-1)$, $B_q^2(k-1)$ is the radius of each neuron in the hidden layer, and $\mathbf{O}^{(2)}(k) \in \mathbb{R}^q$ is the output vector of the hidden layer. The calculation of the output layer is described as (32), $N_r^{(3)}(k)$ is the input of each neuron in the output layer, $\mathbf{W}_r(k-1) \in \mathbb{R}^q$ is the connective weight vector of each neuron of the output layer, which contains the weights from each neuron q to the hidden layer as (34) described, $O_r^{(3)}(k)$ is the output of each neuron in the output layer, $f^{(3)}$ is the active function of the output layer, and Sigmoid function is chosen as the active function with constant gain $C_r \in \mathbb{R}$, and $\mathbf{O}^{(3)}(k) \in \mathbb{R}^3$ is the output vector of the output layer.

Input layer:

$$\begin{cases} N_p^{(1)}(k) = I_p(k) \\ O_p^{(1)}(k) = f^{(1)}(N_p^{(1)}(k)) = N_p^{(1)}(k) \\ \mathbf{O}^{(1)}(k) = [O_1^{(1)}(k), O_2^{(1)}(k), O_3^{(1)}(k)]^T \end{cases} \quad (30)$$

Hidden layer:

$$\begin{cases} N_q^{(2)}(k) = \mathbf{O}^{(1)}(k) \\ O_q^{(2)}(k) = f^{(2)}(N_q^{(2)}(k)) = \exp\left(-\frac{\|N_q^{(2)}(k) - \mathbf{C}_q(k-1)\|^2}{B_q^2(k-1)}\right) \\ \mathbf{O}^{(2)}(k) = [O_1^{(2)}(k), \dots, O_q^{(2)}(k)]^T \end{cases} \quad (31)$$

Output layer:

$$\begin{cases} N_r^{(3)}(k) = \mathbf{W}_r(k-1)\mathbf{O}^{(2)}(k) \\ O_r^{(3)}(k) = f^{(3)}(N_r^{(3)}(k)) = \frac{C_r}{1 + e^{-N_r^{(3)}(k)}} \\ \mathbf{O}^{(3)}(k) = [O_1^{(3)}(k), O_2^{(3)}(k), O_3^{(3)}(k)]^T \end{cases} \quad (32)$$

with

$$\mathbf{C}_q(k-1) = [C_{q1}(k-1), C_{q2}(k-1), C_{q3}(k-1)]^T \quad (33)$$

$$\mathbf{W}_r(k-1) = [W_{r1}(k-1), \dots, W_{rq}(k-1)] \quad (34)$$

The input vector and output vector of the proposed neural network are $\mathbf{I}(k)$ and $\mathbf{K}(k)$ respectively, and the PID gains can be obtained through (35). Note that the inputs of the neural network cannot be obtained at the start of the train running, therefore, initial PID gains of the proposed controller should be set, and the initial value can use gains of the PID controller.

$$\mathbf{K}(k) = \mathbf{O}^{(3)}(k) \quad (35)$$

Remark 7: Poggio and Girosi [32] have proven that the neural network with radial basis function is superior in approximation of continuous function than that with only weight function for its excellent ability of local approximation. Thus, radial basis function is referred in the hidden layer of proposed neural network. It also should be noticed that, the active function of

the output layer is introduced to avoid the oscillation caused by the sign change of the PID gains, the details are provided later.

B. Tuning Parameters of Neural Network

This subsection describes the online tuning method of the neural network parameters through plant I/O data. The network indicator function of error is $E(k)$ shown as (36).

$$E(k) = \frac{1}{2}e^2(k) \quad (36)$$

To tune parameters of neural network, the backpropagation algorithm combined with learning parameter is applied. Each element of $\mathbf{W}_r(k-1)$ is denoted by $W_{rq}(k-1)$, and each element of $\mathbf{C}_q(k-1)$ is denoted by $C_{qp}(k-1)$. Parameter update laws of the proposed neural network are described as (37) of the output layer and (38) of the hidden layer.

$$\begin{cases} W_{rq}(k) = W_{rq}(k-1) + \Delta W_{rq}(k) + \alpha(W_{rq}(k-1) - W_{rq}(k-2)) \\ \Delta W_{rq}(k) = -\eta \frac{\partial E(k)}{\partial W_{rq}(k-1)} \end{cases} \quad (37)$$

$$\begin{cases} C_{qp}(k) = C_{qp}(k-1) + \Delta C_{qp}(k) + \alpha(C_{qp}(k-1) - C_{qp}(k-2)) \\ \Delta C_{qp}(k) = -\eta \frac{\partial E(k)}{\partial C_{qp}(k-1)} \\ B_q(k) = B_q(k-1) + \Delta B_q(k) + \alpha(B_q(k-1) - B_q(k-2)) \\ \Delta B_q(k) = -\eta \frac{\partial E(k)}{\partial B_q(k-1)} \end{cases} \quad (38)$$

where, η is learning rate ($\eta > 0$), α is momentum factor ($\alpha \in [0, 1)$).

Based on the definition of the partial derivative, we have:

$$\begin{aligned} & \frac{\partial E(k)}{\partial W_{rq}(k-1)} \\ &= \frac{\partial E(k)}{\partial y(k)} \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial O_r^{(3)}(k)} \frac{\partial O_r^{(3)}(k)}{\partial N_r^{(3)}(k)} \frac{\partial N_r^{(3)}(k)}{\partial W_{rq}(k-1)} \end{aligned} \quad (39)$$

$$\begin{aligned} & \frac{\partial E(k)}{\partial C_{qp}(k-1)} \\ &= \frac{\partial E(k)}{\partial y(k)} \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial O_r^{(3)}(k)} \frac{\partial O_r^{(3)}(k)}{\partial N_r^{(3)}(k)} \frac{\partial N_r^{(3)}(k)}{\partial O_q^{(2)}(k)} \frac{\partial O_q^{(2)}(k)}{\partial C_{qp}(k-1)} \end{aligned} \quad (40)$$

$$\begin{aligned} & \frac{\partial E(k)}{\partial B_q(k-1)} \\ &= \frac{\partial E(k)}{\partial y(k)} \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial O_r^{(3)}(k)} \frac{\partial O_r^{(3)}(k)}{\partial N_r^{(3)}(k)} \frac{\partial N_r^{(3)}(k)}{\partial O_q^{(2)}(k)} \frac{\partial O_q^{(2)}(k)}{\partial B_q(k-1)} \end{aligned} \quad (41)$$

The partial derivative of relevant segments can be referred in (42)–(49). Note that the Jacobian information $\partial y(k)/\partial u(k)$ is hard to get for the complicated plant, thus, the estimated value is used to obtain this value as (43), and error caused by this adjustment can be made up by a proper learning rate [8]. The partial derivatives of PID output control signal $u(k)$ to the PID gains are displayed in (44). The calculation of remaining formulas are

deduced from the proposed neural network described as (31) and (32).

$$\frac{\partial E(k)}{\partial y(k)} = -e(k) \quad (42)$$

$$\frac{\partial y(k)}{\partial u(k)} \approx \frac{y(k) - y(k-1)}{u(k) - u(k-1)} \quad (43)$$

$$\frac{\partial u(k)}{\partial O_r^{(3)}(k)} = \begin{cases} e(k), r = 1 \\ \sum_{i=0}^k e(i) \Delta t, r = 2 \\ (e(k) - e(k-1))/\Delta t, r = 3 \end{cases} \quad (44)$$

$$\frac{\partial O_r^{(3)}(k)}{\partial N_r^{(3)}(k)} = \frac{C_r e^{-x}}{(1 + e^{-x})^2} \Big|_{x=N_r^{(3)}(k)} \quad (45)$$

$$\frac{\partial N_r^{(3)}(k)}{\partial W_{rq}(k-1)} = O_q^{(2)}(k) \quad (46)$$

$$\frac{\partial N_r^{(3)}(k)}{\partial O_q^{(2)}(k)} = W_{rq}(k-1) \quad (47)$$

$$\frac{\partial O_q^{(2)}(k)}{\partial C_{qp}(k-1)} = O_q^{(2)}(k) \frac{x - C_{qp}(k-1)}{B_q^2(k-1)} \Big|_{x=N_q^{(2)}(k)} \quad (48)$$

$$\frac{\partial O_q^{(2)}(k)}{\partial B_q(k-1)} = O_q^{(2)}(k) \frac{\|x - \mathbf{C}_q(k-1)\|^2}{B_q^3(k-1)} \Big|_{x=N_q^{(2)}(k)} \quad (49)$$

C. PID Gain Analysis

According to the functionalities of PID gains, usually, signs of gains are the same for a specific plant [11]. The change of gain sign may cause oscillation of PID controller output, which implies the opposite action for the plant. For the constructed train model, if K_p turns to be negative with positive tracking error, the controller outputs will decrease and further enlarge the tracking error. This situation exists similarly in K_i and K_d .

PID gains are the outputs of proposed neural network, and the proposed neural network without active function $f^{(3)}$ of the output layer is discussed in this subsection. In order to distinguish the proposed neural network with and without $f^{(3)}$, NN and NN' are used respectively. The momentum factor α is set as 0 in this subsection to simplify update laws.

Without the active function of the hidden layer, the output layer of NN' is shown as (50), and $\mathbf{W}_r(k-1)$ of NN' is updated through (51) and (52) with $\alpha = 0$.

$$\begin{cases} N_r^{(3)}(k) = \mathbf{W}_r(k-1) \mathbf{O}^{(2)}(k) \\ O_r^{(3)}(k) = N_r^{(3)}(k) \\ \mathbf{O}^{(3)}(k) = [O_1^{(3)}(k), O_2^{(3)}(k), O_3^{(3)}(k)]^T \end{cases} \quad (50)$$

$$\begin{cases} W_{rq}(k) = W_{rq}(k-1) + \Delta W_{rq}(k) \\ \Delta W_{rq}(k) = -\eta \frac{\partial E(k)}{\partial W_{rq}(k-1)} \end{cases} \quad (51)$$

$$\frac{\partial E(k)}{\partial W_{rq}(k-1)} = \frac{\partial E(k)}{\partial y(k)} \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial O_r^{(3)}(k)} \frac{\partial N_r^{(3)}(k)}{\partial W_{rq}(k-1)} \quad (52)$$

Then the neuron output of the output layer $O_r^{(3)}(k+1)$ can be obtained through (53) for NN'. Note that Gaussian function is the active function of the hidden layer, which implies the hidden layer outputs are positive. Thus, the sign of $O_r^{(3)}(k+1)$ of NN' is decided by $\mathbf{W}_r(k)$ as (54) shown. For the various situations, the sign of $\mathbf{W}_r(k)$ may change. So the oscillation caused by the gain sign change cannot be eliminated through the design of NN'. And the negative PID gain has been indeed observed in the test of NN' with the constructed train model.

$$O_r^{(3)}(k+1) = \mathbf{W}_r(k) \mathbf{O}^{(2)}(k+1) \quad (53)$$

$$\begin{aligned} \mathbf{W}_r(k) &= \mathbf{W}_r(k-1) + \eta e(k) \frac{y(k) - y(k-1)}{u(k) - u(k-1)} \\ &\quad \times \frac{\partial u(k)}{\partial O_r^{(3)}(k)} \mathbf{O}^{(2)}(k) \end{aligned} \quad (54)$$

Theorem 1: In the proposed neural network NN, using Sigmoid function as the active function of the output layer ensures the invariant sign of PID gain.

Proof: For NN, the neuron output of the output layer $O_r^{(3)}(k+1)$ can be expressed as (55). The active function of the hidden layer is Gaussian function, and the outputs of the hidden layer are always positive. The $\mathbf{W}_r(k)$ can be calculated through (56) with $\alpha = 0$ for NN. Even though the sign of $\mathbf{W}_r(k)$ may change, the Sigmoid function of (55) will ensure the sign of $O_r^{(3)}(k+1)$ invariant.

$$O_r^{(3)}(k+1) = \frac{C_r}{1 + e^{-x}} \Big|_{x=\mathbf{W}_r(k) \mathbf{O}^{(2)}(k+1)} \quad (55)$$

$$\begin{aligned} \mathbf{W}_r(k) &= \mathbf{W}_r(k-1) \\ &+ \eta e(k) \frac{y(k) - y(k-1)}{u(k) - u(k-1)} \frac{C_r e^{-N_r^{(3)}(k)}}{(1 + e^{-N_r^{(3)}(k)})^2} \\ &\quad \frac{\partial u(k)}{\partial O_r^{(3)}(k)} \mathbf{O}^{(2)}(k) \end{aligned} \quad (56)$$

Remark 8: The PID gains are positive for the constructed train model of this study. The default value of C_r is 1 for the proposed controller, and the value can be adjusted if needed. For other plants, the PID gains may be negative, and this situation can be realized with negative C_r .

D. Strategy Under Actuator Saturation

As many studies illustrated, actuator saturation can result in instability of the system, and this phenomena is severer in most adaptive control systems [19], [28]. In this study, the proposed controller without saturation strategy is disabled under continuous saturated output. From (39), (40), and (41), it can be seen that $\partial y(k)/\partial u(k)$ is used in parameter tuning laws of the proposed controller. If the actuator is saturated with upper bound (or lower bound) of sampling sequence k and $k+1$, then the value of denominator of (43) can be 0 through (57). The denominator of $\partial y(k)/\partial u(k)$ with zero value is meaningless and will cause

system breakdown.

$$u(k) - u(k-1) = \begin{cases} \bar{U} - \bar{U} = 0, U(k) > \bar{U}, U(k-1) > \bar{U} \\ \underline{U} - \underline{U} = 0, U(k) < \underline{U}, U(k-1) < \underline{U} \end{cases} \quad (57)$$

To avoid the disability of parameter tuning under actuator saturation, which means $u(k) \neq U(k)$. The parameter tuning strategy (58) is used under this circumstance.

$$\begin{cases} W_{rq}(k) = W_{rq}(k-1) \\ C_{qp}(k) = C_{qp}(k-1) \\ B_q(k) = B_q(k-1) \end{cases} \quad (58)$$

Assumption 1: If $u(k) = \bar{U}$ and $u(k-1) < \bar{U}$, then $e(k) > 0$, $\Delta e(k+i) < 0$, with $1 \leq i \leq k_c$, $u(k+i-1) = \bar{U}$, $k_c \in \mathbb{N}^+$. If $u(k) = \underline{U}$ and $u(k-1) > \underline{U}$, then $e(k) < 0$, $\Delta e(k+i) > 0$, with $1 \leq i \leq k_c$, $u(k+i-1) = \underline{U}$, $k_c \in \mathbb{N}^+$.

Remark 9: The condition of *Assumption 1* exactly means two conditions, $u(k) = \bar{U}$ while train outputs the maximum tracking force, and $u(k) = \underline{U}$ while train outputs the maximum braking force. In practice, train outputs upper bound traction signal with a big positive tracking error, while lower bound with a big negative traction error. And the maximum traction/braking ability of the train is sufficient to satisfy the general operation requirements. Thus, the maximum traction/braking force can hold extremely running situations, which implies the absolute value of running error is decreased under this situation.

Theorem 2: Train controller can get out from saturation with parameter tuning method (58). Specifically, when the actuator is saturated, if $u(k) = \bar{U}$, then there exist k_s that can make $u(k+k_s) < \bar{U}$; if $u(k) = \underline{U}$, then there exist k_s that can make $u(k+k_s) > \underline{U}$.

Proof: For upper saturation, the controller output under saturation can be described as following:

$$\begin{aligned} U(k+k_x) &= K_p(k+k_x)e(k+k_x) + K_i(k+k_x)\Delta t \sum_{i=0}^{k+k_x} e(i) \\ &\quad + K_d(k) \frac{e(k+k_x) - e(k+k_x-1)}{\Delta t} \end{aligned} \quad (59)$$

where, $k+k_x$ is the sampling sequence under saturation with $1 \leq i \leq k_x$, $u(k+i) = \bar{U}$, $k_x \in \mathbb{N}^+$. $K_p(k+k_x)$, $K_i(k+k_x)$, and $K_d(k+k_x)$ are all positive values because of the Sigmoid function combined with positive C_r of the output layer in proposed neural network. From *Assumption 1*, it can be known that $e(k+k_x)$ and $e(k+k_x) - e(k+k_x-1)$ must shrink constantly under saturation. Thus, the increase tendency of $\Delta t(e(0) + e(1) + \dots + e(k+k_x))$ must be covered at a specific sampling sequence k_s , which results in the decrease of $U(k+k_s)$. For lower saturation, the conclusion can be obtained through similar method.

IV. SIMULATION STUDY

Simulation is a common method to validate train running with small costs and risks. To show the effectiveness of the proposed NNPID controller for subway trains, relevant controllers

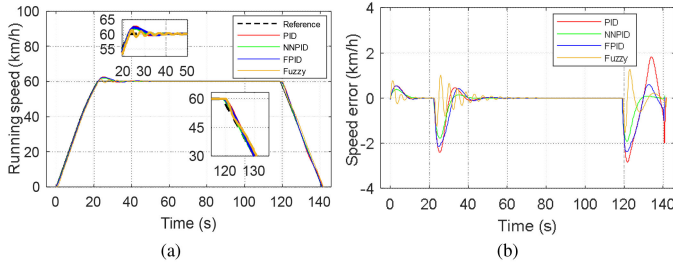


Fig. 4. Speed tracking curve versus time of scenario 1.

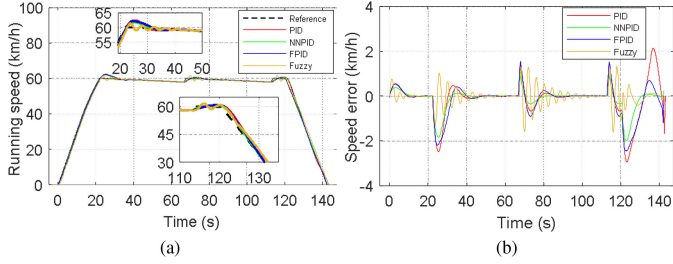


Fig. 5. Speed tracking curve versus time of scenario 2.

are compared. Besides, the proposed controller is applied to a subway line, and tracking performance is analyzed.

A. Numerical Tests and Validation

To illustrate the characteristics of the proposed controller, numerical tests are made for the subway train. PID controller, fuzzy controller, and FPID controller are constructed to make comparisons under the same condition, and the fuzzy rules are modified based on relevant studies [15], [21]. The fuzzy logic of fuzzy controller outputs plant control signal, while the fuzzy logic of FPID controller decides PID gains. The initial values (K_p , K_i , K_d) of PID, FPID and NNPID are the same as (0.6, 0.3, 0.2).

The simulations are carried out with reference speed profile in different scenarios to verify the control performance of the proposed NNPID controller. The speed profile is usually designed with three typical strategies, which are cruising strategy, coasting strategy, and cruising plus coasting strategy [3]. The cruising strategy of scenario 1 is based on the driving regimes such as acceleration, cruising, and braking. The coasting strategy of scenario 2 is based on the driving regimes such as acceleration, cycles of cruising and reacceleration, and braking. The cruising plus coasting strategy of scenario 3 is based on the driving regimes such as acceleration, cruising, coasting and braking. For each scenario, the parameters of reference signal and dynamic train model are the same for different controllers. The running section for numerical tests is 2 km of length with flat track, the total running time of reference speed profiles are 141 s, 142.5 s, and 141.5 s for three scenarios.

To show the tracking performance of different speed profiles, figures of speed tracking and speed error are displayed in Fig. 4, for scenario 1, Fig. 5 for scenario 2, and Fig. 6 for scenario 3. Fig. 4(a), Fig. 5(a), and Fig. 6(a) show that NNPID controller performs best in comprehensive results of overshoot and transition

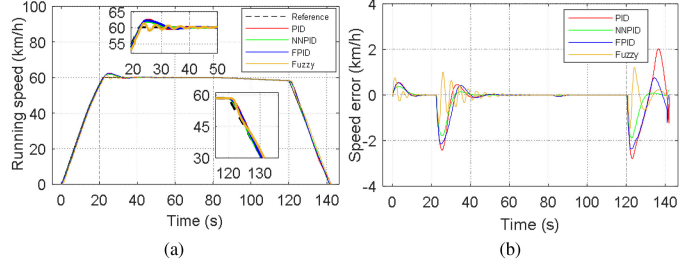


Fig. 6. Speed tracking curve versus time of scenario 3.

TABLE I
RESULTS OF NUMERICAL TEST

Scenario	Item	PID controller	NNPID controller	FPID controller	Fuzzy controller
1	Running time (s)	141.50	141.00	140.90	141.80
	Stopping position (m)	2000.22	2000.02	2000.09	2000.03
	Energy (kWh)	14.56	14.16	14.40	15.70
2	Running time (s)	143.30	142.50	142.40	143.20
	Stopping position (m)	2000.19	2000.02	2000.11	2000.03
	Energy (kWh)	15.28	14.59	15.42	20.85
3	Running time (s)	142.40	141.50	141.50	142.20
	Stopping position (m)	2000.12	2000.02	2000.11	2000.03
	Energy (kWh)	13.92	13.54	13.79	15.10

time. In Fig. 4(b), Fig. 5(b), and Fig. 6(b), the NNPID controller has visible smaller error variances than other controllers in each scenario, which means a more stable control.

Relevant results of train running are shown as Table I. The results of NNPID controller are the nearest values of reference speed profile in running time and stopping position. It is also should be noted that stopping positions of all controllers satisfy the error requirement (0.3 m). As for energy consumption, NNPID controller still performs best, and fuzzy controller is the worst one. This phenomena is largely caused by system fluctuations.

For the difference in plant models, the controllers may have different performances from the studies we referred. And fuzzy logic relies on human experience, which means the fuzzy logic should be set carefully. It is noteworthy that the tracking precision may be improved if PID gains and fuzzy logic are adjusted through more trials. Compared with the trivial and tricky adjusting, NNPID controller with automatic parameter tuning is easier to utilize.

B. Application Case Study

In this subsection, the parameters are borrowed from Beijing Subway Yizhuang Line for the case study. There are 14 stations and 13 running sections with 22.728 km long.

The formulas of train traction characteristic curve and braking characteristic curve are shown as (60) and (61). The unit of traction/braking force is kN.

$$f_{a\max}(\dot{x}(t)) = \begin{cases} 310, \dot{x}(t) \leq 36 \text{ km/h} \\ 0.074\dot{x}^2(t) - 11.84\dot{x}(t) + 640.62, & (60) \\ 36 < \dot{x}(t) \leq 80 \text{ km/h} \end{cases}$$

$$f_{b\max}(\dot{x}(t)) = \begin{cases} 260, \dot{x}(t) \leq 60 \text{ km/h} \\ 560 - 5\dot{x}(t), 60 < \dot{x}(t) \leq 80 \text{ km/h} & (61) \end{cases}$$

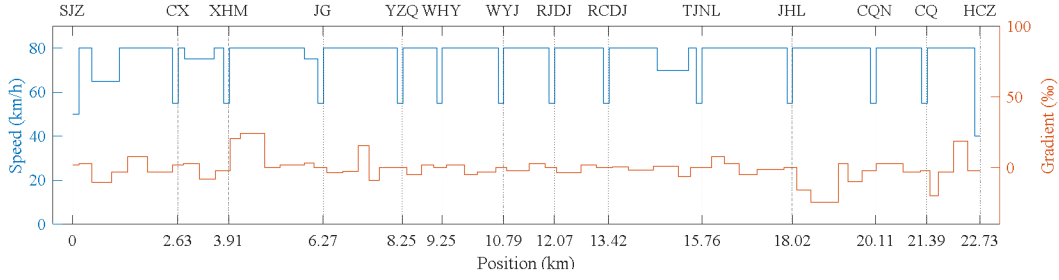


Fig. 7. Speed limit, gradient and station position of Yizhuang line.

TABLE II
TRAIN PARAMETERS OF BEIJING SUBWAY YIZHUANG LINE

Parameters	Value	Conditions
$a_u(t)$	2.031 daN/t	$4800 \leq x(t) < 18136$ m
	3.063 daN/t	$0 \leq x(t) < 4800$ m, $18136 \leq x(t) \leq 22728$ m
$b_u(t)$	0.0622 daN/t* h /km	$4800 \leq x(t) < 18136$ m
	0.06484 daN/t* h /km	$0 \leq x(t) < 4800$ m, $18136 \leq x(t) \leq 22728$ m
$c_u(t)$	0.001807 daN/t* h^2 /km ²	$4800 \leq x(t) < 18136$ m
	0.008213 daN/t* h^2 /km ²	$0 \leq x(t) < 4800$ m, $18136 \leq x(t) \leq 22728$ m

TABLE III
TRAIN MASS OF OFF-PEAK HOURS

Section No.	Running section	Passenger mass (kg)	Train mass $M(t)$ (t)
1	SJZ-XC	46170	244.17
2	XC-XHM	43740	241.74
3	XHM-JG	42090	240.09
4	JG-YZQ	35040	233.04
5	YZQ-WHY	31890	229.89
6	WHY-WYJ	30120	228.12
7	WYJ-RJDJ	25890	223.89
8	RJDJ-RCDJ	23610	221.61
9	RCDJ-TJNL	17460	215.46
10	TJNL-JHL	8790	206.79
11	JHL-CQN	5940	203.94
12	CQN-CQ	5790	203.79
13	CQ-HCZ	1680	199.68

* Station name abbreviation: SJZ (Songjiazhuang), CX (Xiaocun), XHM (Xiaohongmen), JG (Jiugong), YZQ (Yizhuangqiao), WHY (Wenhuayuan), WYJ (Wanyuanjie), RJDJ (Rongjingdongjie), RCDJ (Rongchangdongjie), TJNL (Tongjinanlu), JHL (Jinghailu), CQN (Ciqunan), CQ (Ciqu), HCZ (Huochezhan).

As for basic resistance per unit mass, the parameters are different for ground and underground. The underground resistance is larger than that of the ground due to the effect of tunnel. Table II shows relevant parameters of the basic resistance per unit mass.

Different running section leads to different passenger load. The net mass of the train is 198 t, and the passenger mass data with off-peak hours used in the simulation is shown as Table III.

The data of speed limit, gradient, and station position can be referred in Fig. 7. The gradient $\iota(t)$ is decided by the train running position. Besides, the uncertain disturbance of unit mass $\tilde{\epsilon}_u(t)$ is estimated through the normal distribution with 10 s holding, $\tilde{\epsilon}_u(t) = h(t) - h(t - 10)$, $h(t) \sim N(0, 0.017^2)$.

As for the parameters of proposed neural network, learning rate η is 0.5, momentum factor α is 0.05.

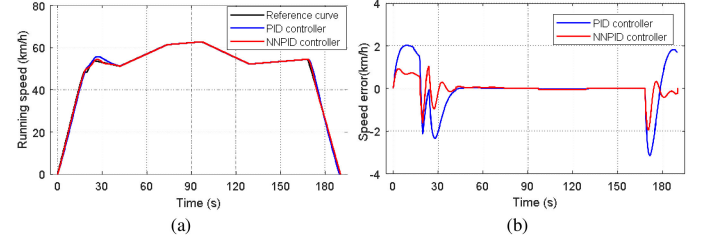


Fig. 8. Speed tracking curve versus running time of section SJZ-XC.

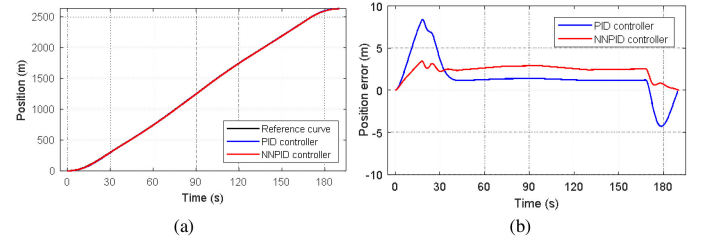


Fig. 9. Position tracking curve versus running time of section SJZ-XC.

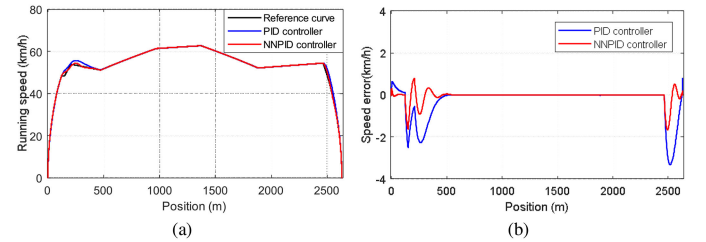


Fig. 10. Running speed versus position of section SJZ-XC.

To show the performance of proposed controller clearly, the detail result figures of the first running section SJZ-XC are displayed. Fig. 8 shows the speed tracking curve versus running time of section SJZ-XC, Fig. 9 shows the position tracking curve versus running time of section SJZ-XC, and Fig. 10 shows the speed tracking curve versus train running position of section SJZ-XC, which can be obtained through Fig. 8 and Fig. 9. Besides, the train traction/braking force, control signal, and acceleration versus running position of section SJZ-XC are displayed in Fig. 11.

From the curves in Figs. 8–10, it can be seen that the proposed NNPID controller performs well in reference speed

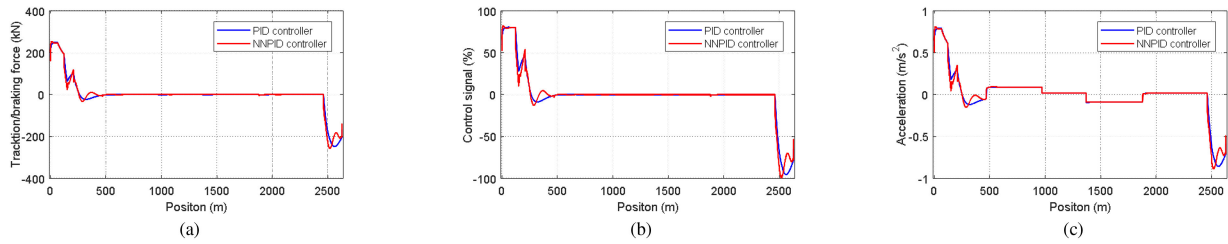


Fig. 11. Traction/braking force, control signal, and acceleration versus position of section SJZ-XC.

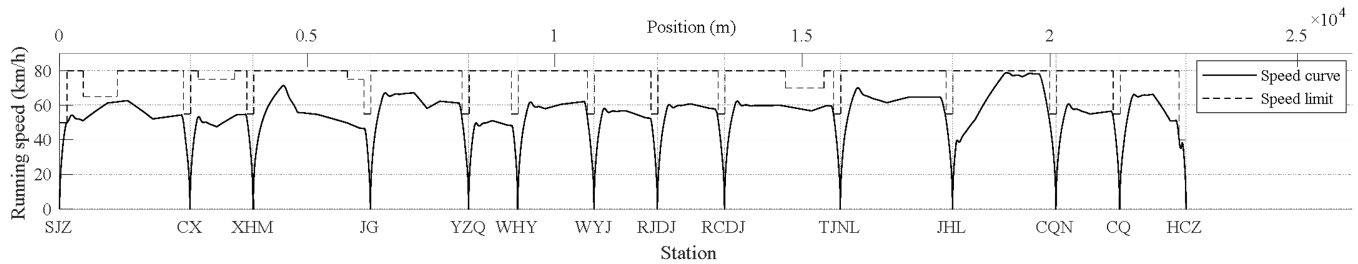


Fig. 12. Train running speed curve of the Beijing Subway Yizhuang Line in down direction.

TABLE IV
DETAILED RUNNING INDEXES OF THE CONTROLLERS FOR YIZHUANG LINE IN DOWN DIRECTION

Section No.	Running section	Running time error (s)		Stopping position error (m)		Maximum absolute speed error (m/s)		Energy (kWh)	
		PID controller	NNPID controller	PID controller	NNPID controller	PID controller	NNPID controller	PID controller	NNPID controller
1	SJZ-XC	0.70	-0.10	0.04	0.02	0.93	0.47	10.78	10.46
2	XC-XHM	0.20	-0.30	-0.21	-0.17	1.83	0.60	10.23	10.21
3	XHM-JG	1.00	0.10	-0.23	0.02	2.22	0.50	28.74	28.36
4	JG-YZQ	1.20	-0.20	-0.07	0.03	1.40	0.53	14.98	14.72
5	YZQ-WHY	0.40	-0.21	-0.27	-0.18	2.49	0.94	8.52	8.54
6	WHY-WYJ	1.00	0.00	-0.26	0.05	2.27	0.53	13.47	13.20
7	WYJ-RJDJ	0.90	0.00	-0.31	-0.22	2.61	0.70	11.50	11.23
8	RJDJ-RCDJ	1.20	-0.10	-0.14	0.02	1.85	0.44	12.07	11.84
9	RCDJ-TJNL	0.70	0.00	-0.18	-0.25	2.84	0.75	13.52	13.27
10	TJNL-JHL	1.10	-0.10	-0.29	0.01	0.86	0.45	17.65	17.33
11	JHL-CQN	0.80	0.00	0.02	0.01	0.89	0.49	5.85	5.98
12	CQN-CQ	0.20	-0.40	-0.17	-0.28	3.82	0.71	13.36	13.10
13	CQ-HCZ	1.00	0.20	-0.38	-0.16	2.19	3.34	11.64	11.33

profile tracking. Tracking error variances of speed versus time, position versus time, and speed versus position of the proposed NNPID controller are all less than classic PID controller, which means the proposed NNPID controller performs better under the same running situations. Note that each error is calculated under the same horizontal axis of involved curves, and some errors are not equal to 0 at the end of curves for the different running time and stopping position. Fig. 11 provides the detailed information in traction/braking force, control signal, and acceleration value. The control signal is the percent of the maximum force at each running speed. And acceleration curve in Fig. 11 implies the train acceleration fulfills the requirement that the subway train acceleration value should be less than 1 m/s².

To obtain the overall performance, the proposed controller is applied to the whole line. The running speed profiles of all sections of Yizhuang line in down direction is displayed as Fig. 12. What can be known from Fig. 12 is that the proposed controller can control the train with daily operation requirements.

There are four indicators used to evaluate the tracking performance of the line, specifically are running time error, stopping position error, maximum absolute speed error, and energy. Running time error and stopping position error are the differences between reference speed profile and practice running in total running time and stopping position respectively. Running time error and stopping running error measure the deviations of total running time and train stopping position. The maximum absolute speed error is the maximum absolute value of difference between reference and practice running speed, this indicator shows the speed tracking performance of the controller. Energy indicates the energy consumed by motors during running, and energy consumption is an optimal objective of many train control studies [3].

Table IV represents detailed running data of the whole line, the comparison of currently used PID controller shows that the proposed controller has better performance. The proposed controller performs better in majority indexes of each running section compared with classic PID controller. Note that running

TABLE V
SUMMARIZED RUNNING INDEXES OF THE CONTROLLERS FOR YIZHUANG LINE IN DOWN DIRECTION

Data	Average absolute error of running time (s)	Average absolute error of stopping position (m)	Average maximum absolute error of speed (m/s)	Total energy consumption (kWh)
PID	0.8	0.2	2.02	172.31
NNPID	0.13	0.11	0.8	169.57
NNPID-PID	-0.67	-0.09	-1.22	-2.74
(NNPID-PID)/PID	-83.75%	-45%	-60.4%	-1.5%

time error and stopping position error for each running section of both controllers satisfy the requirements.

Table V shows the overview optimization with average indicators. Compared with PID controller, the average values of absolute running time error, absolute stopping error, and absolute maximum speed error of proposed NNPID controller are improved with 0.67 s, 0.09 m and 1.22 m/s, which means 83.75%, 45%, and 60.4% better. Besides, the total energy consumption of the line in down direction can save 2.74 kWh with 1.5% improvement under the same conditions.

This study proposed an adaptive NNPID controller for subway train running control, and PID gains as well as actuator saturation are taken into account. Through the real line application, it can be found that the NNPID controller is easy to use for the simple structure, and the controller performance is good. Compared with PID controller, NNPID controller can adaptively tune PID gains based on the real time information, which is ensured by the information processing of neural network. Similar to the analysis at the end of last subsection, it also should be admitted that the performance of PID controllers may be improved through more tuning work of PID gains.

V. CONCLUSION

To optimize the train control effect of subway train, an adaptive controller based on PID control is proposed. The main contributions of this paper are the train model with dynamic parameters as well as a NNPID controller, which inherits the main structure of PID controller.

Aiming to adjust PID gains adaptively and efficiently, a neural network is constructed based on Gaussian function of the hidden layer, and the characteristics of PID gains and actuator saturation are considered. Based on the train model with dynamic parameters, the effectiveness of the proposed NNPID controller is demonstrated through numerical tests and real line application. Compared with PID controller, FPID controller, and fuzzy controller under the same conditions, the proposed controller performs best. As for the application of Beijing Subway Yizhuang Line, the proposed NNPID controller outperforms the currently used PID controller. And for the whole line, the average values of absolute running time error, absolute stopping error, and absolute maximum speed error are improved with 0.67 s, 0.09 m and 1.22 m/s respectively. The total energy is also decreased with 2.74 kWh of one train in single down direction running. The results implies better tracking effect and less energy consumption of the proposed NNPID controller, and this benefit will be amplified with the application of all trains.

There are multiple trains running on the line at the same time, and the running of the trains influence each other. The proposed

train running controller of this study focuses on the single train, which means limited optimization. In the future, multiple train running control will be taken into account of our study to make a comprehensive optimization.

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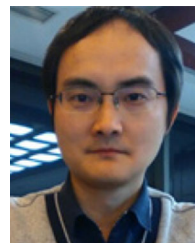
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